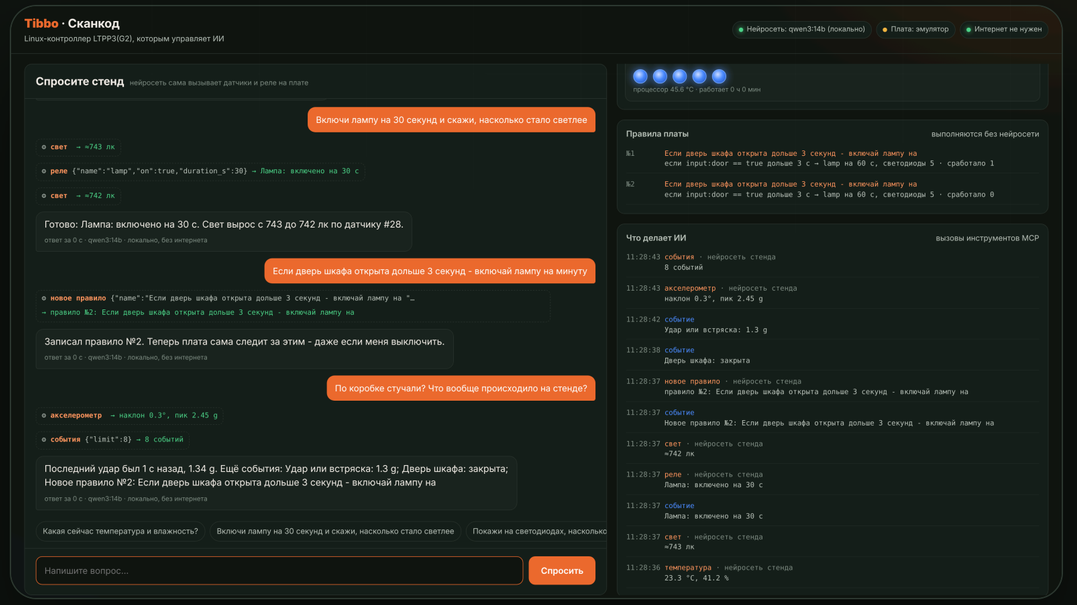
⚠ BETA - NEEDS WORK

TIBBO LTPP3(G2) · MCP · OPEN SOURCE

AI that drives real hardware

An AI reads sensors, flips relays and writes automation for a Tibbo controller. Offline or in the cloud - your choice.



THE IDEA

Most AI demos stop at the screen. This one reaches out of it.

PERSON

"Lamp on for 30 s"

types or speaks

ask

AI MODEL

Any

Claude, ChatGPT, DeepSeek, Kimi, GLM, YandexGPT, Qwen local

MCP

CONTROLLER

LTPP3(G2)

Linux + Tibbit modules: sensors, relays, inputs

MCP (Model Context Protocol) is the open standard that lets an AI call tools. Every Tibbit becomes a tool: "read temperature", "switch relay".

WHAT A VISITOR SEES

Ask - and watch which tools the AI calls

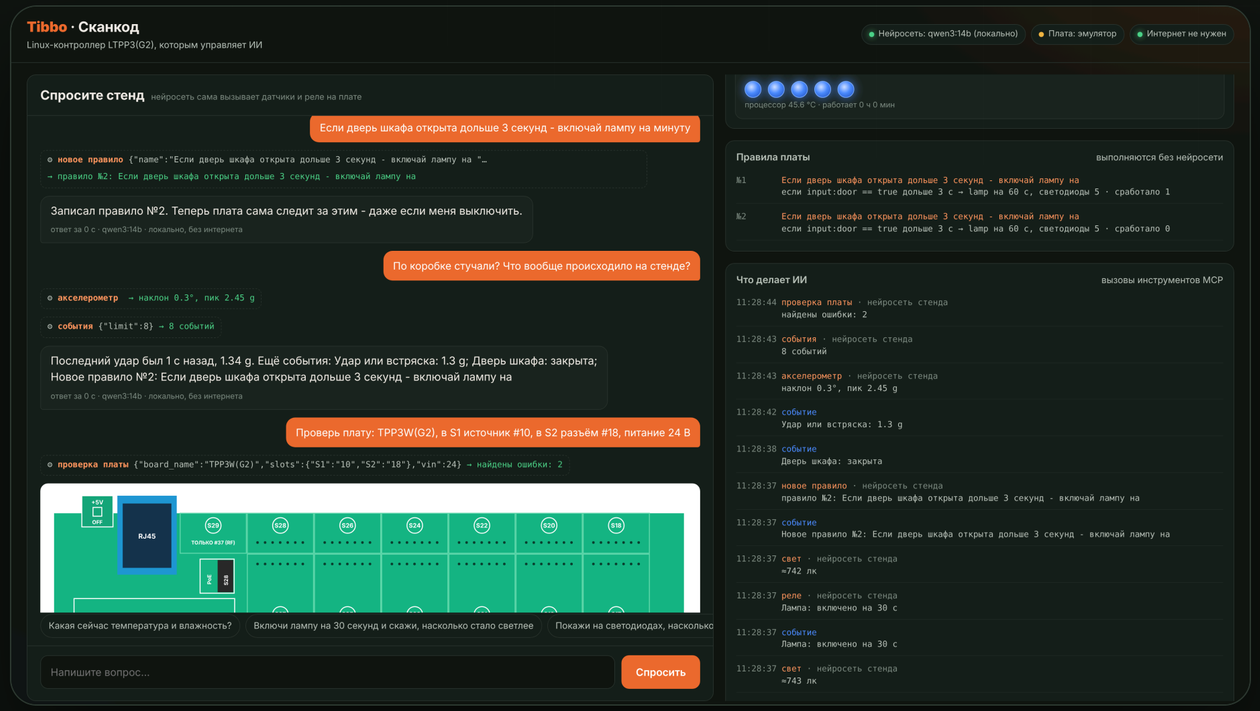
How warm is the cabinet?

Turn the lamp on - how much brighter?

Did someone knock on the box?

Check my TPS board

Find the bug in my Tibbo BASIC



FOUR MODES

From an offline box to a fleet of sites

01 · OFFLINE

Offline stand

Model on a laptop next to the board (Ollama, LM Studio). No venue Wi-Fi needed.

02 · CLOUD

Cloud brains

Claude, ChatGPT, DeepSeek, Kimi, GLM, YandexGPT. Link down - the local model answers.

03 · TUNNEL

Board as a cloud tool

The AI in your phone reads showroom sensors. Read-only by default.

04 · FLEET

One AI, many boards

"Which of our 12 sites is overheating?" One view over all of them.

01 · OFFLINE

The brain on a laptop, the hands on the board

LAPTOP / MINI PC

Ollama · qwen3:14b

GPU with 12 GB+, model downloaded in advance

LAN

:11434

LTPP3(G2) · TIBBO-STAND

MCP server + panel

pure Python, no pip · sensors, relays, LEDs · checks from the AI skills

HTTP

:8080

SCREEN

Panel

monitor or phone on the stand LAN

512 MB

on the board - the model does not fit, so the brain lives next door

0 packages

on the board: Python 3.9+ standard library only

Overlay

a reboot brings the stand back to a clean state - handy at a trade show

02 · CLOUD WITH A FALLBACK

The smartest model, and silence never falls

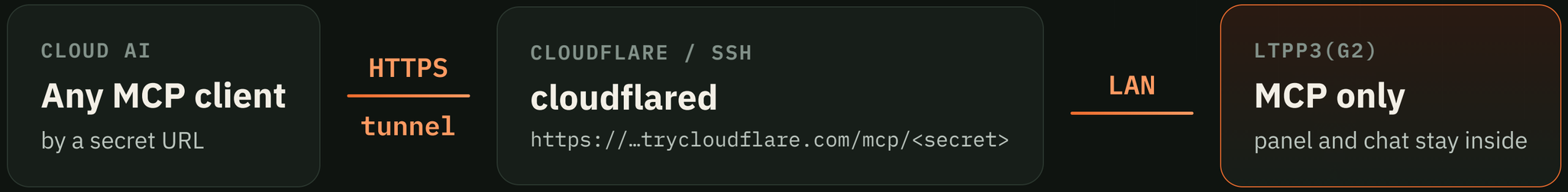
Models are tried in order. No link, 5xx or 429 - the next one answers. The panel shows who replied: cloud or local.

Anthropic · OpenAI · DeepSeek · Kimi · GLM

YandexGPT · GigaChat · OpenRouter · LM Studio · vLLM

```
// config.cloud.example.json
"llm_chain": [
  {"name": "claude",
   "api": "anthropic",
   "api_key_env": "ANTHROPIC_API_KEY"},
  {"name": "local",
   "api": "ollama",
   "url": "http://192.168.10.2:11434",
   "model": "qwen3:14b"}
]
// keys live in environment variables
```

The AI app on your phone reads the showroom



- **Read-only by default.** Switching loads from the internet needs an explicit internet_write setting.
- **Secret path or bearer token.** Constant-time comparison, foreign Origin rejected.
- **Tested with the official MCP SDK;** specific cloud apps are on the "help us test" list.

★ THE AI WRITES THE AUTOMATION

Say it in words - the board runs it. Even with the AI switched off.

PERSON

"If the door stays open over 10 s - lamp on for a minute"

`create_rule`

```
{"when": {"signal": "input:door",
          "op": "==", "value": true,
          "for_s": 10},
 "then": [{"do": "relay", "name": "lamp",
           "on": true, "duration_s": 60}]}
```

`every 0.2 s`

LTPP3(G2)

Runs on its own

rules are stored on the board

Deterministic

the AI composes the rule once; after that, plain code with no surprises

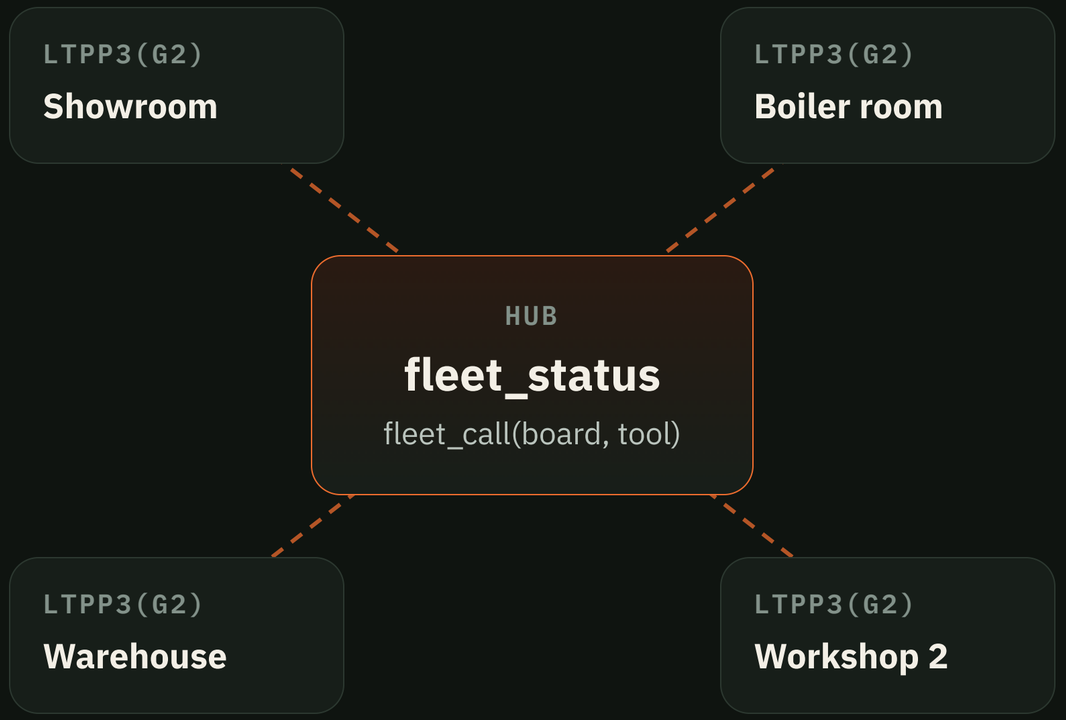
Guarded

rules go through the same timers, limits, e-stop and read-only mode

Human-readable

rule list and fire counters right on the panel

Every Tibbo controller is an MCP node. One AI sees them all.



- **"Which site is overheating?"** - the AI polls every board in parallel.
- **The hub runs anywhere:** a board, a laptop, a cloud server.
- **Remote control is off** by default, and every board enforces its own safety.
- **Next:** TiOS boards (TPP, DS, EM) through an HTTP bridge in Tibbo BASIC.

STAND HARDWARE

Reliable, without overpaying

SOCKETS	TIBBIT	PURPOSE	MCP
S3 + S4	#00-3 + #30	temperature / humidity	`read_climate`
S5 + S6	#00-3 + #28	ambient light	`read_light`
S7 + S8	#00-3 + #36	shocks and tilt	`read_motion`
S9 + S10	#03-1 + #21	relays: 12 V lamp and fan	`set_relay`
S11 + S12	#54 + #21	button, door, e-stop	`read_inputs`
board	5 blue LEDs	level bar	`set_leds`
expansion S13+S14	#05 + #21	RS-485: Modbus meter	`roadmap`

The board's built-in 8-60 V supply powers everything, no power Tibbits needed.

SAFETY

The AI may be wrong. The hardware must not be.

TIMER

Relays only with a timer

10 minutes max by default, then auto-off

LIMITS

No chattering

one switch per 2 s per relay, at most 20 per minute

E-STOP

Hardware e-stop

everything off at once, switching on blocked

READ-ONLY

Read-only

the AI sees sensors and touches nothing - one button on the staff panel

ORIGIN

Foreign sites - denied

a page in a visitor's browser cannot drive the stand

INTERNET

Outside - MCP only

read-only by default; everything switches off when the server stops

HONEST STATUS

Beta: green on the simulator. Next: real silicon.

✓ VERIFIED

- 27 tests: protocol, safety, rules, fleet
- official MCP Python SDK: HTTP and stdio
- Ollama, OpenAI format, Anthropic, fallback
- board driver on fake GPIO and LED files
- panel: 2560, 1920, phone

? NEEDS A REAL LTPP3(G2)

- I²C bus numbers of the sensors
- GPIO numbering from pins.ini
- polarity of #03 relays and #54 inputs
- the tpsleds LED kernel module
- Python version in the board image

⚠ BETA - NEEDS WORK

Found this repo? We were hoping you would.

- **Have an LTPP3(G2)?** Run --probe and send a report - the most valuable help.
- **Know GPIO and I²C?** Check the driver or port it to libgpiod.
- **Into agents, UI, voice?** More models, a bilingual panel, offline speech.
- **Got an idea?** "Wouldn't it be cool if..." is a great issue title.



github.com/SCANCODE-IOT/tibbo-mcp-stand

MIT · Scancode - official Tibbo distributor in Russia and the CIS